MiroEval: Benchmarking Multimodal Deep Research Agents in Process and Outcome

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Research Questions

- How can we comprehensively evaluate deep research systems along both the quality of final outputs and the underlying research processes?
- What are the relative strengths and weaknesses of current deep research systems across synthesis quality, factual accuracy, and process rigor?
- How do multimodal capabilities affect the performance of deep research systems compared to text-only tasks?

Core Idea

Deep research systems generate long-form reports by conducting multi-step investigations, but existing benchmarks only evaluate final outputs using fixed rubrics, missing crucial aspects like research process quality and multimodal handling. MiroEval introduces a three-dimensional evaluation framework: adaptive synthesis quality (task-specific rubrics), agentic factuality verification (claims verified against web and multimodal sources), and process-centric evaluation (auditing search, reasoning, and refinement steps). The key insight is that process quality predicts outcome quality while revealing invisible weaknesses like traceability gaps between research steps and final claims.

Key Formula

Inverse Quality Filter

`Q_{gen} = \{q | \sigma(q) \leq 0.75 \land \ell(q) \neq \text{high} \land \text{requires\_search}(q)\}`

σ(q) is continuous quality score [0,1] for baseline parametric answer, ℓ(q) is categorical quality label, requires_search(q) is binary flag. Retains only queries where parametric knowledge baseline is demonstrably inadequate, ensuring benchmark tasks require external investigation.

Process Score

$$S_{process} = \alpha S_{intrinsic}(P) + (1-\alpha) S_{align}(P, R)$$

P is process record, R is final report, S_intrinsic measures intrinsic process quality across five dimensions, S_align measures bidirectional alignment between process findings and report content. Combines process capability assessment with traceability measurement.

Key Figure

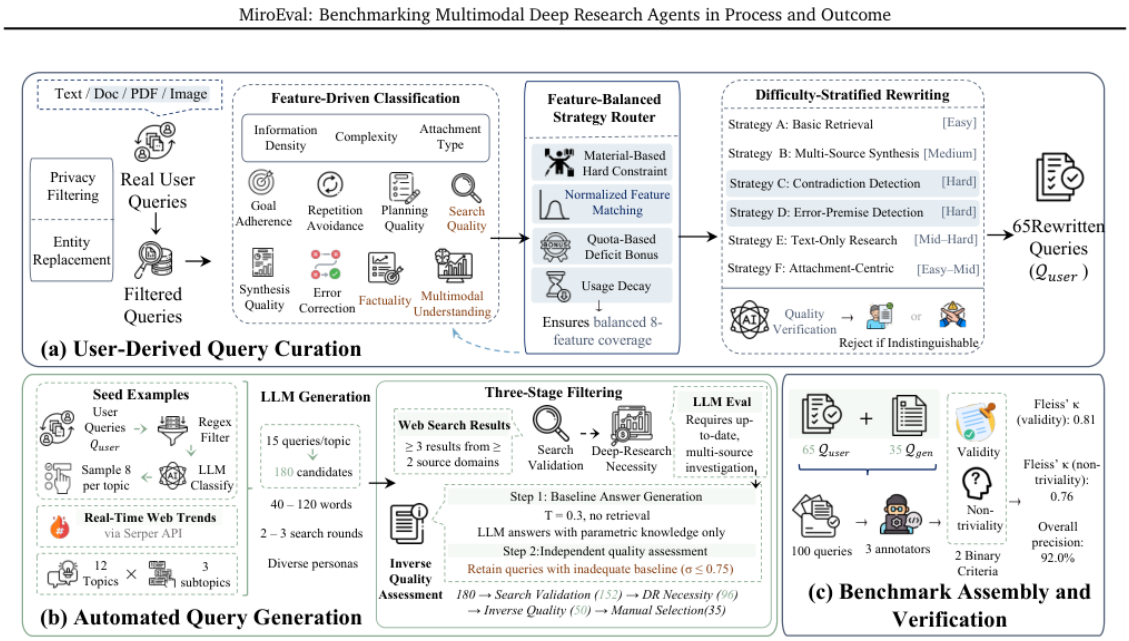

Figure 2: Overview of query construction pipeline.

Figure 2: Shows the dual-path benchmark construction pipeline combining user-derived query curation with automated trend-grounded generation, enabling both real-world grounding and scalable refresh capability.

Limitations

- Process evaluation requires systems to expose intermediate reasoning traces, limiting applicability to closed-source systems without such access
- Factuality evaluation identifies cross-source conflicts but does not resolve them - CONFLICT label flags disagreements without determining correct source
- Current implementation focuses on English tasks and may not generalize to other languages or cultural contexts

- Benchmark size of 100 tasks may be insufficient for fine-grained analysis of specialized domains or rare task types